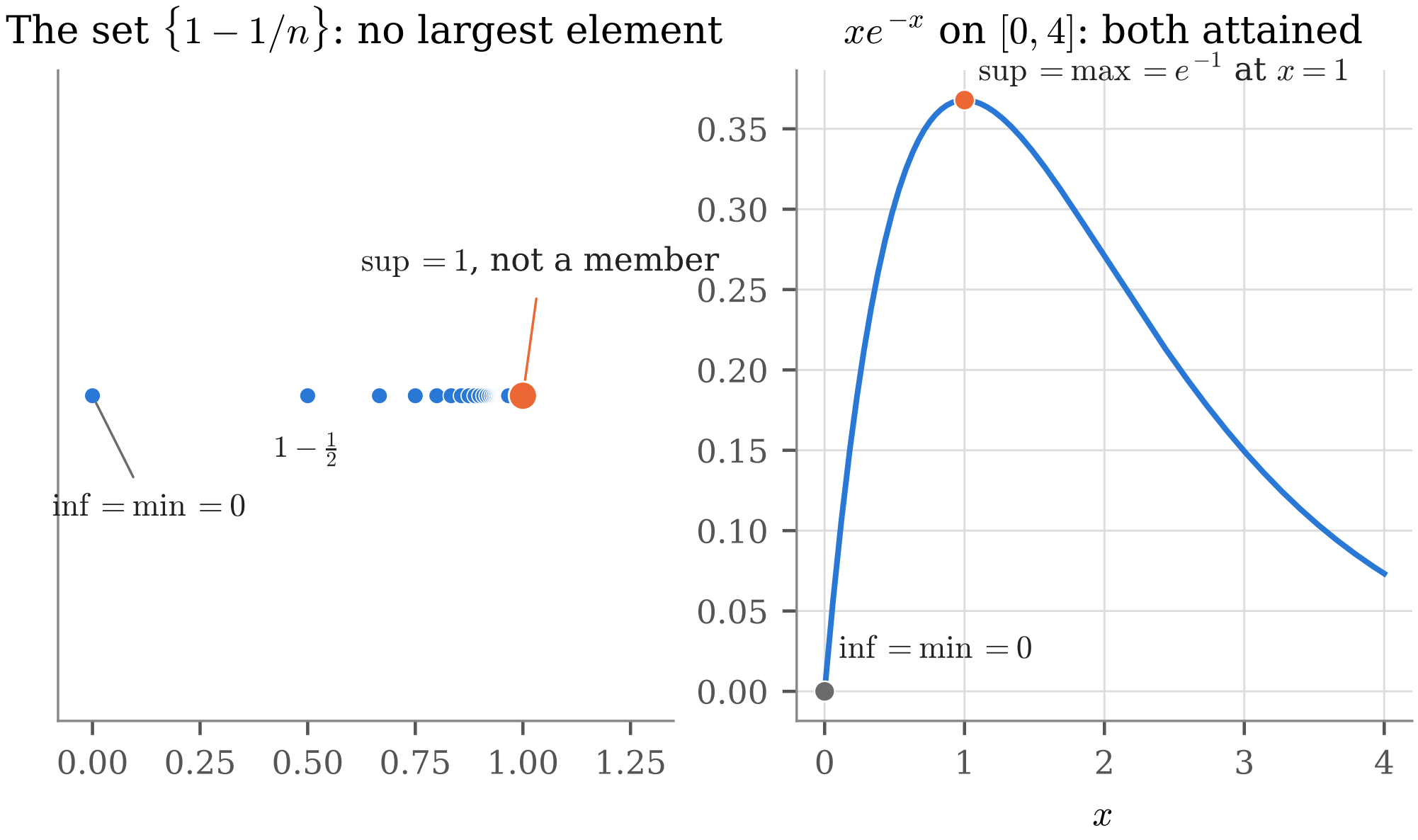
The set $\{1 - 1/n\}$: no largest element
$\sup = 1$, not a member
$1 - \frac{1}{2}$
$\inf = \min = 0$
0.00
0.25
0.50
0.75
1.00
1.25
xe^{-x} on $[0, 4]$: both attained
$\sup = \max = e^{-1}$ at $x = 1$
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
$\inf = \min = 0$
0
1
2
3
4
x